PAKISTAN
STOCK EXCHANGE

KSE100 INDEX

(Total Return & Price Return)

INDEX BROCHURE COVERING INDEX METHODOLOGY

(UPDATED: JUNE 2025)

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1. INTRODUCTION

The KSE 100 Index is designed to measure the performance of 100 companies which comprises the sector largest market capitalization companies and highest market capitalization companies (other than sector largest market capitalization companies). This index has two variants - Total Return Index and Price Return Index. The KSE100 Price Return Index (KSE100PR) is designed to measure the price performance of same KSE100 companies listed on the Pakistan Stock Exchange (PSX). The price variant only reflects the capital appreciation of index constituents and does not account for the impact of reinvestment of gross cash dividends.

The KSE-100 Index was introduced in November 1991 with base value of 1,000 points. The Index comprises of 100 companies selected on the basis of sector representation and highest free-float market capitalization. The Price Return version of the index began from 01 April 2009 (back dated). The KSE100 indices captures around ~70% to 80% of the total free-float market capitalization of the all the companies listed on the Exchange. Out of all the sectors listed on the Exchange, top company from each sector (excluding Open-End Mutual Fund and Exchange Traded Funds) on the basis of the largest free-float market capitalization is selected. To make the total 100 constituents of the index, the remaining companies are selected on the basis of largest free-float market capitalization in descending order.

The KSE100 index is a total return index i.e. dividend, bonus, rights and any other corporate actions are adjusted in the constituents of the index, whereas, the KSE100PR Index is a price return index where gross cash dividends are not adjusted in the constituents of the index.

Index Expert Committee (IEC) of PSX recommended to the governing board of directors of the Pakistan Stock Exchange Limited (PSX) in early 2012 to implement the KSE-100 Index on the basis of free float market capitalization. In the meeting held on April 24, 2012, the governing PSX Board approved the IEC recommendation. The free float based KSE-100 index was calculated parallel to the full-cap KSE-100 Index since June 11, 2012 and the recomposed KSE100 Index based on free float methodology effective from October 15, 2012. In this transition, the rules for composition and re-composition of the Index based on free float methodology have remained unchanged other than selection of companies on the basis of free-float market capitalization as against total market capitalization.

2. OBJECTIVE

The primary objective of the KSE100 index is to have a benchmark by which the stock price performance can be compared to over a period of time. In particular, the KSE100 index is designed to provide investors with a sense of how the Pakistan's equity market is performing. Thus, the KSE100 index is similar to other indicators that track various sectors of the Pakistan's economic activity such as the gross national product, consumer price index, etc. The KSE100PR index is designed to provide investors with a sense of how the KSE100 constituents are performing in terms of price change. Thus, the KSE100PR index is same in terms of methodology to the KSE100 index with the exception of not adjusting gross cash dividends in the Price return variant.

3. FREE FLOAT METHODOLOGY

Free float means proportion of total shares issued by a company that are readily available for trading at the stock exchange. It generally excludes the shares held by controlling directors / sponsors / promoters, government and other locked-in shares not available for trading in the normal course.

3.1. Objective and Description

- Free Float calculation can be used to construct stock indices for better market representation than those constructed on the basis of total market capitalization of companies.
- It gives weight for constituent companies as per their actual liquidity in the market and is not unduly influenced by tightly held large-cap companies.
- Free Float can be used by the Exchange for regulatory purposes such as risk management and market surveillance.

Free Float Calculation Methodology			
Total Outstanding Shares			**Xxx**
Less	Government Holdings	xxx	
	Shares held by Directors / Sponsors / Senior Management Officers and their Associates	xxx	
	Shares in Physical Form	xxx	
	Shares held by Associate Companies / Group Companies (Cross Holdings)	xxx	
	Shares issued under Employees Stock Option Scheme that cannot be sold in the Open market in normal course	xxx	
	Treasury Shares	xxx	
	Any other category that are barred from selling at the review date	xxx	
Free Float			**Xxx**

Notwithstanding to the above calculations, under no circumstances, free float of a scrip shall exceed its book entry shares, available in the Central Depository System.

'Sponsor' has the same meaning as defined in The Companies (Issue of Capital) Rules, 1996

'Senior Management Officer' and 'Associate' have the same meaning as defined in the Securities Act, 2015

4. SELECTION CRITERIA OF STOCKS

- Largest Free-Float market capitalization in each of the *36* Pakistan Stock Exchange sectors *excluding Exchange Traded Funds* and *Open-end Mutual Funds sector*;

List of sectors of PSX are as follows:

List of Sectors			
1	Automobile Assembler	20	Oil & Gas Exploration Companies
2	Automobile Parts & Accessories	21	Oil & Gas Marketing Companies
3	Cable & Electrical Goods	22	Open-end Mutual Funds *(excluded)*
4	Cement	23	Paper, Board & Packaging
5	Chemical	24	Pharmaceuticals
6	Close-end Mutual Funds	25	Power Generation & Distribution
7	Commercial Banks	26	Property
8	Engineering	27	Real Estate Investment Trust
9	Exchange Traded Funds *(excluded)*	28	Refinery
10	Fertilizer	29	Sugar & Allied Industries
11	Food & Personal Care Products	30	Synthetic & Rayon
12	Glass & Ceramics	31	Technology & Communication
13	Insurance	32	Textile Composite
14	Investment Bank/ Inv. Cos/ Sec. Cos.	33	Textile Spinning
15	Jute	34	Textile Weaving
16	Leasing Companies	35	Tobacco
17	Leather & Tanneries	36	Transport
18	Miscellaneous	37	Vanaspati & Allied Industries
19	Modaraba	38	Woollen

- The remaining index places (*currently 64*) are taken up by the largest free float market capitalization companies in descending order.
- Company which is on the Defaulters' Counter and/or its trading is suspended, declared Non-Tradable (i.e. NT) in preceding six months from the date of re-composition shall not be considered in the re-composition of KSE-100 Index.
- A newly listed company qualifies to be included in the existing index (after one re-composition period) if the free float capitalization of the newly listed company is at least 2% of the total free float market capitalization

5. RE-COMPOSITION / REBALANCING

KSE100 Index & KSE100PR Index shall be recomposed on semi-annual basis as follows:

DATA BASIS	NOTICE*	IMPLEMENTATION
Last working day of February	15th of March	First working day of April
Last working day of August	15th of September	First working day of October

*If that is a holiday, the notice will be issued on the next working day

6. ADDITION & DELETION OF STOCKS DURING RE-COMOPOSITION PROCESS

Maintenance of the two indices over time will require an on-going semi-annual re-composition process, internal and external- buffer files of shares that exceed (shares outside the index) or fall below (shares inside the index) the above criteria will be maintained under the jurisdiction of the Board of Directors/Management of the Exchange.

Maintaining adequate representation of the underlying constituents of the stock market through all of its future development and changes is dependent upon the establishment of an appropriate re-composition process.

Re-composition rules fall into two general categories:

1) Sector Rules and

2) Free Float Market Capitalization Rule.

6.1 Sector Rules

Sector rules govern the selection (or deletion) of companies on the basis of being the top free float market capitalization stock from each of the PSX sectors (excluding Open-end Mutual Fund and Exchange Traded Fund sector). Two rules are recommended to undertake in selection under this, a time-based rule and the other is a value-based rule. Stocks shall be triggered by compliance with either rule.

Time-based rule:

A stock (not in the index) which becomes the largest in its sector (by any amount of value) shall be made part of the index, if that company is able to maintain its position as the largest in the sector for consecutively two re-composition periods.

Value-based rule:

A stock (not in the index) which becomes the largest in its sector by a minimum of 10% greater in free-float market capitalization value than the present largest in the sector (in the index) will enter the index at the time of re-composition.

6.2 Capitalization Rule

Capitalization rule governs the selection (or deletion) of companies on the basis of being among the largest free float market capitalization companies in the stock market. Only one rule applies here that is a time-based rule.

Time-based rule:

A stock (not in the index) may qualify for entry if it exceeds the free float market capitalization value consecutively in two re-composition periods of the last stock in the index selected on the basis of market cap. A qualifying company automatically pushes out the lowest cap selected stock in the index.

6.3 Special Re-composition

The exchange has the right to do special re-composition in the case of some extra ordinary events like voluntary delisting, scheme of arrangement, merger, etc. The exchange will intimate all the market participants before implementing this special re-composition through a notice.

7. BASE PERIOD

The base period of KSE100 Index is November 1991 and the base value is 1,000 index points. The calculation of KSE100 Index involves dividing the free float market capitalization of 100 companies in the Index by a number called the Index Divisor. The base period of KSE100PR index is April 01, 2009 and the base value is 6,931 index points. Similarly, the calculation of KSE100PR index involves dividing the free float market capitalization of 100 companies in the Index by a number called the Index Divisor. The divisor is the only link to the original base period value of the KSE 100 Indexes. It will keep the index comparable over a period of time and will also be the adjustment point for all future corporate actions, replacement of scrips, etc.

8. MAINTENANCE OF KSE100 INDEX

The day-to-day maintenance of the index will be carried out within the Broad Index Policy Framework set out by the Exchange. The management will ensure that KSE100 Index and KSE100PR Index and all the other PSX indexes maintain their benchmark properties by striking a balance between periodic re-composition and rebalancing of the indexes and maintaining their historical continuity.

9. ON-LINE COMPUTATION OF INDEX

During market hours, the prices of all the underlying securities at which trades are executed, are automatically used by the trading computer to calculate the KSE100 and KSE100PR index and continuously make updates on all trading workstations connected through PSX trading terminals on real time basis.

10. ADJUSTMENT IN THE KSE100 INDEX FOR CORPORATE ACTIONS

KSE100 Index shall be calculated on total return basis and corporate actions (Cash dividend, Bonus and Right issue) adjustments shall be made in order to maintain the index continuity. KSE100PR index shall be calculated on price return basis and all corporate actions adjustments shall be made except gross cash dividend in order to maintain the index continuity. If no adjustments were made, a discontinuity would arise between the current value of the index and its previous value despite the non-occurrence of any economic activity of substance. At the Exchange, the base value will be adjusted, which is used to alter market capitalization of the component stocks to arrive at the KSE100 Index value.

The determination of Ex-price of a security is mentioned in Rule 10.6 of PSX Rule Book that: “If the Books of a Security are closed for determining any entitlement for its shareholders by the Issuer, the Exchange shall determine the ex-price based on the mechanism prescribed by the Exchange, as an opening price for the Trading Day falling on two Settlement Day before its Books Closure start date”.

The key difference in the adjustment of corporate actions between the two variants of the KSE100 & KSE100PR index lies in how they account for cash dividend:

Impact	KSE100PR	KSE100
Cash Dividends	No	Yes
Bonus Adjustment	Yes	Yes
Right Adjustment	Yes	Yes

New Divisor shall be calculated due to corporate action at the end of T-3 days of its Book Closure starting date. E.g. Starting day of Book Closure = Friday, new divisor shall be calculated at day end of Tuesday.

The adjustment for corporate actions will be made as given under:

10.1 Adjustment for Cash Dividend (only for KSE100 Index)
KSE100 Index is a total return index; therefore, adjustment of cash dividend shall be made. However, no adjustment will be made for KSE100PR index for cash dividend.

If company declared a 10% cash dividend:

KSE100 Index = 1,120 points

KSE100 Index market capitalization = Rs. 13,950,000,000

Divisor = 12,455,357

Constituents	Share Price	Free Float Shares	Free Float Market Capitalization
Stock A	22.50	50,000,000	1,125,000,000
Stock B	41.00	150,000,000	6,150,000,000
Stock C	44.50	150,000,000	6,675,000,000
Total Free Float Market Capitalization			**13,950,000,000**

Step-1

Determine the ex-dividend price of the stock A to calculate the revised market capitalization and a new divisor for the next day

Stock A

Par value: Rs.10 per share

Closing Price: Rs. 22.50 per share

Cash Dividend 10 % of par value

i) Cash dividend amount per share = par value x dividend% = Rs 10 x 10% = Rs.1

ii) Ex-dividend price = closing price – cash dividend amount = Rs.22.50 – Rs.1 = Rs.21.50

Step-2

Illustration, Revision in the Market Capitalization and Divisor

Share price of stock A shall be adjusted after the close of day to calculate the New Divisor for the next day

Constituents	Share Price	Free Float Shares	Free Float Market Capitalization
Stock A	21.50	50,000,000	1,075,000,000
Stock B	41.00	150,000,000	6,150,000,000
Stock C	44.50	150,000,000	6,675,000,000
Revised Free Float Market Capitalization			**13,900,000,000**

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New Divisor = Revised Market Cap. / Index points

13,900,000,000 / 1,120 = 12,410,714

10.2 Adjustment for Bonus Shares (for both variants)

Declaration of Bonus requires adjustment in the free float capitalization and within the index divisor itself. The following process illustrates the process for a situation whereby a stock A has declared 10% bonus in terms of shares. Following steps are recommended to be followed in order to determine the price of Ex-Bonus of Stock A to calculate the revised free float market capitalizations and new divisor for the next day.

Step-1

KSE100 Index = 1,120 points

KSE 100 Index Free-float market capitalization = Rs. 13,950,000,000

Divisor= 12,455,357

Closing Price of Stock A: Rs. 22.50

Bonus: 10 %

Stock lot size= 100 Shares

Given below example, calculates the Ex-bonus price on the basis of a stock lot size of 100 shares by observing following steps.

- Total free-float shares after the Bonus issue: 100 + (100 X 10 %) = 110 shares
- Cost of stock-lot size 100 shares x closing price of stock A: 100 x 22.50= Rs. 2,250

  Ex-Bonus Price: 2,250/110 = RS 20.45

Step-2

Illustration, Revision in the Market Capitalization and Divisor

Share price and free-float shares of stock A shall be adjusted after the close of day to calculate the New Divisor for the next day

Stock A		
Free Float shares	**Bonus**	**Total free float shares**
50,000,000	10%	55,000,000

Constituents	Share Price	Free Float Shares	Free Float Market Capitalization
Stock A	20.45	55,000,000	1,124,750,000
Stock B	41.00	150,000,000	6,150,000,000
Stock C	44.50	150,000,000	6,675,000,000
Revised Free Float Market Capitalization			**13,949,750,000**

*New Divisor = Revised Market Cap. / Index points

13,949,750,000 / 1,120 = 12,455,134

**Divisor changed as stock prices are reported in two decimal places*

10.3 Adjustment for Right Shares (for both variants)

The Right issues of the companies which constitute the KSE100 Index are adjusted in two stages. At first stage the Ex-Right price is adjusted and at the second stage the capital (free-float shares) are adjusted. A brief detail about the right issues is mentioned below:

The company which declares Right shares has to close its books (shareholders register) to determine entitlement within 30 days of its declaration.

At the date of book closure, the Ex–Right price is ascertained and if the company belongs to the KSE 100 Index then the Divisor is adjusted due to the Ex-Right price of the company.

When the company informs the Exchange that it has dispatched Letter of Rights Offer to the shareholders, the trading in the Letter of Rights Offer (Un- paid) are commenced. A separate block of capital, Un-Paid-Right, is formed equal to amount of right issue and the trading continues till next 30 days or till the last date of payment.

After the last date of payment, the trading in Un-Paid-Right (Letter of Rights Offer) is discontinued.

By the end of 30th day of the last date of payment or earlier, the company informs that share certificates are ready for exchange with Right Allotment Letter (RAL) or credited in the CDS, the capital of the RAL is merged with the company. At this stage the Divisor of the KSE100 Index is adjusted for the increase in the number of shares of the company.

A) Right issue without premium

If Company A has issued 10 % right shares

KSE100 Index = 1,120

KSE100 Index Free Float Market Capitalization = 13,950,000,000

Divisor = 12,455,357

First Stage

Step 1

Determine the Ex-Right price of the stock A to calculate the revised free-float market capitalization and a new divisor for the next day.

Stock A price Rs 22.50

Right: 10 %

For simplicity in working, we will calculate the Ex-Right price on the basis of a lot of 100 shares.

i. Total free-float shares after the Right issue

100 shares + (100 shares x 10 % Right) = 110 shares

ii. Cost of a lot (100 shares)

(100 shares x market price of A) + (10 right shares x par value)

= 100 x 22.50+ 10 x 10

= Rs. 2350

iii. Ex- Right price per share = 2,350/110

= Rs. 21.36

Step 2

Share price of A is adjusted after the close of day to calculate the New Divisor for the next day.

Stock	Shares Price (Rs)	No of Free Float Shares	Market Value (Rs)
A	21.36	50,000,000	1,068,000,000
B	41	150,000,000	6,150,000,000
C	44.5	150,000,000	6,675,000,000
Revised Free Float Market Capitalization			**13,893,000,000**

New Divisor = Revised Free Float Market Capitalization/ Index points

=13,893,000,000/1,120 = 12,404,464

Second Stage

After 15 days of the last date of payment the company confirm the subscription amount, accordingly the capital of RAL is merged with the company and the Divisor is adjusted for the increase in number of free-float shares.

Step 1

Free-float shares of Stock A shall be adjusted after the close of day to calculate the New Divisor for the next day, assuming that the closing price of stock A is at 22

Stock A		
Free Float shares	**Right Issue**	**Total Free Float shares**
50,000,000	10%	55,000,000

Stock	Shares Price (Rs)	No of Free Float Shares	Market Value (Rs)
A	22	55,000,000	1,210,000,000
B	41	150,000,000	6,150,000,000
C	44.5	150,000,000	6,675,000,000
Revised Free Float Market Capitalization			**14,035,000,000**

New Divisor = Revised Free Float Market Capitalization/ Index Points

New Divisor = 14,035,000,000/1,120 =12,531,250

More details related to corporate action adjustment methodology is available on the following link:

https://www.psx.com.pk/psx/themes/psx/uploads/priceCalMethod.pdf

DISCLAIMER

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